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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: 25-26J-148

Project Title: Automatic Cricket Video Segmentation By Player Identity And Event Type

	Identification and Analysis
Data Privacy Concerns	The cricket videos used in this research are publicly available cricket match videos, such as videos from YouTube. Therefore, the privacy risk from the video content is low because the players and match events are already publicly visible. The system does not collect sensitive personal information such as addresses, contact numbers, financial data, or private player data. It only extracts cricket-related information such as player names shown on the scoreboard, batsman/bowler roles, event types, and timestamps for academic video segmentation and retrieval. However, if the website includes user accounts, basic user information such as username, email address, and login details should be handled carefully and used only for authentication and system access purposes.
Data Security	The system stores uploaded cricket videos, processed video segments, extracted scoreboard text, player-event records, timestamp details, and website user account details. Even though cricket video data is not highly sensitive, website user accounts must be protected properly. User passwords should not be stored as plain text; they should be securely hashed before storing. Login pages should use authentication, secure session/token handling, and proper logout functions. Backend APIs and dashboard pages should be protected so that only authorized users can upload videos, view match details, and access processed results. File uploads should be validated to allow only safe video files, and unnecessary temporary files should be removed after processing. If deployed online, HTTPS should be used to protect login details and data transmission.
Bias and Fairness	The system may not give the same level of accuracy for all cricket videos because results depend on video quality, scoreboard clarity, camera angle, lighting, and umpire visibility. For example, OCR may misread a player name if the scoreboard is blurry, and the event model may misclassify an umpire signal if the umpire is not clearly visible. This may lead to wrong event-player matching. Therefore, the system results should be treated as automatically generated outputs and should be verified before using them for serious coaching or player evaluation. User account access should also be fair, meaning authorized users should have appropriate access based on their role and should not be able to view or modify data outside their permission level.

Transparency and Accountability	The project clearly explains the main processing steps used in the system, including scoreboard OCR, NER-supported player identification, fuzzy matching, umpire-signal event detection, timestamp synchronization, and dashboard-based retrieval. The system does not claim perfect accuracy because OCR, NER, and event classification can make mistakes. The methodology, limitations, accuracy values, and possible errors are documented as part of the research. Users of the dashboard can understand that the displayed results are generated by the system and may require manual checking when used for serious analysis. The user account feature is also included only to control access to the system and protect the uploaded and processed data.
Impact on Stakeholders	The main stakeholders of this research are cricket players, coaches, analysts, researchers, viewers, video/content owners, and website users. Coaches and analysts benefit because the system reduces manual video review and helps them find player-specific events more quickly. Researchers benefit because the project demonstrates a practical multimodal sports video analysis system. Players are not directly harmed because the videos used are publicly available and the system only extracts cricket-related information. However, wrong predictions could affect the quality of analysis, so the results should be checked before being used for important performance decisions. Website users benefit from account-based access, while their login information and account details are protected within the system. Content owners are also considered because publicly available YouTube videos may still have copyright restrictions.